EXAMPLE 2

Hellenic Republic of Greece Annual Coupon Bond

At the height of the COVID pandemic, the government of Greece issued a 2 percent annual coupon bond maturing in seven years.

1. If the observed YTM at issuance was 2.00 percent, what was the issuance price (PV) per EUR100 of principal?

   Solution:

   EUR100
   Solve for PV using Equation 6 with PMT_i = EUR2, r = 2.00%, and FV = EUR100:

   $$PV = \text{EUR}100$$

   $$= \frac{2}{1.02} + \frac{2}{1.02^2} + \frac{2}{1.02^3} + \frac{2}{1.02^4} + \frac{2}{1.02^5} + \frac{2}{1.02^6} + \frac{2}{1.02^7}.$$

   We may solve this using the Microsoft Excel or Google Sheets PV function introduced earlier (PV (0.02,7,2,100,0)). The issuance price equals the principal ($PV = FV$). This relationship holds on a coupon date for any bond where the fixed periodic coupon is equal to the discount rate.
   The present value of each cash flow may be solved using Equation 5. For example, the final EUR102 interest and principal cash flow in seven years is:

   $$PV = \text{EUR}88.80 = \text{EUR}102/(1.02)^7.$$

   The following table shows the present value of all bond cash flows:

Years/Periods	0	1	2	3	4	5	6	7
FV		2	2	2	2	2	2	102
PV	100.00	1.96	1.92	1.88	1.85	1.81	1.78	88.80

2. Next, let's assume that, exactly one year later, a sharp rise in Eurozone inflation drove the Greek bond's price lower to EUR93.091 (per EUR100 of principal). What would be the implied YTM expected by investors under these new market conditions?

   Solution:

   3.532 percent
   In this case, we must solve for r using Equation 6, with PV equal to 93.091, as follows:

   $$PV = 93.091 = 2/(1+r) + 2/(1+r)^2 + 2/(1+r)^3 + 2/(1+r)^4 + 102/(1+r)^5.$$

   Here we may use the Microsoft Excel or Google Sheets RATE function (`RATE (5,2,93.091,100,0,0.1)`) to solve for r of 3.532 percent.
   Investors in fixed coupon bonds face a capital loss when investors expect a higher YTM.

The interest rate r used to discount all cash flows in Equation 6 is the bond's YTM, which is typically quoted on an annual basis. However, many bonds issued by public or private borrowers pay interest on a semiannual basis. In Example 3, we revisit the Republic of India bond from which a single cash flow was stripped in an earlier simplified example.